

Review article

Digital twins in cardiovascular disease: a scoping review

Huina Zou ^{a,1}, Xinglin Zheng ^{a,b,1}, Linjing Wu ^a, Shujie Zhang ^a, Polun Chang ^{c,*}, Yuan Chen ^{a,*}^a Xiamen Cardiovascular Hospital of Xiamen University, School of Medicine, Xiamen University, Xiamen, China^b School of Nursing, Fujian University of Traditional Chinese Medicine, Fuzhou, China^c Institute of Biomedical Informatics, National Yang-Ming Chiao-Tung University, Taipei, Taiwan

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ABSTRACT

Background: Digital twin technology in healthcare is an emerging approach that creates virtual representations of patients and disease-specific conditions, with the potential to clarify treatment objectives and enable more personalized, precision-based care, to help to clarify treatment objectives and to facilitate personalised and precision treatment management.

Objective: This scoping review was conducted to analyse the application of digital twin technology in cardiovascular disease, focusing on implementation steps, clinical applications, and challenges to guide future research.

Methods: A systematic search was conducted in eight databases (PubMed, EBSCO, Web of Science, WILEY, China WanFang Database, China National Knowledge Infrastructure, China Weipu Database, and SinoMed) for studies published, with a time frame of database construction to May 2025. Data were summarised and analysed based on predefined criteria.

Results: A total of 31 cardiovascular studies were included. Their implementation was categorised into five stages: data acquisition, model construction and personalisation, model calibration and validation, simulation analysis, and result application for decision support or medical education. Clinical applications involved personalised health management (13 %), precise individual treatment effects (42 %), individual risk prediction (26 %), clinical trial optimisation (23 %), and medical education (3 %). Key challenges included data limitations, model construction and validation complexities, and barriers to clinical application.

Conclusion: Digital twins demonstrate potential in cardiovascular care by advancing personalised health management and precision medicine. However, their widespread adoption and practical implementation are still in their early stages. Broader implementation necessitates improved data sharing, algorithm optimisation, enhanced model generalizability, and ethical safeguards.

What was already known on the topic?

- Digital twins originated in industry and have been applied to medicine to facilitate personalized health management.
- Digital twins are effective in diseases such as cancer and diabetes, and are moving from concept to application.
- There are already reviews summarizing the theoretical basis of cardiovascular digital twins, such as proof of concept.
- In cardiovascular disease, which remains the leading global cause of death, there is a paucity of comprehensive, evidence-based syntheses mapping DT applications, technical maturity, and gaps, hindering their translation into routine care.

What this study added to our knowledge

- Proposed a five-stage DT implementation framework (data acquisition, individualized modeling, calibration/validation, simulation/prediction, clinical application/education) applicable across cardiovascular subspecialties.
- Mapped 31 eligible studies into five quantified application domains, involved personalised health management (13 %), precise individual treatment effects (42 %), individual risk prediction (26 %), clinical trial optimisation (23 %), and medical education (3 %).
- Identified cross-cutting challenges in data integration, model validation, and clinical workflow embedding, offering targeted

* Corresponding authors.

E-mail address: 28837445@qq.com (Y. Chen).¹ Huina Zou and Xinglin Zheng contributed equally as the first authors of this study.

recommendations for multi-center, prospective, real-time/closed-loop evaluations to accelerate safe and effective adoption

1. Introduction

Cardiovascular diseases (CVD) continue to be the leading cause of death and disability worldwide [1]. More precise and efficient strategies are required to enhance patient outcomes due to the complexity and variability of CVD [2]. Originally developed for industrial applications, Digital Twin (DT) technology has recently been introduced into healthcare, where it holds transformative potential for personalized and precision medicine [3]. However, there is currently no universally accepted definition of a medical digital twin. The scoping review by Drummond et al. [3] suggested defining a patient digital twin as “a viewable digital replica of a patient, organ, or biological system that contains multidimensional, patient-specific information and informs decisions”. In healthcare, digital twins serve a dual purpose: they can act as data generators to simulate patient conditions, treatment responses, and multiscale biological processes, while also serving as data sources for advanced analytics through artificial intelligence and machine learning, often integrated with retrospective and cohort datasets [4,5]. For instance, virtual models of cardiovascular disease progression can be built using patient cohort data and AI algorithms to predict disease onset and inform the design of personalized cardiac rehabilitation programs. This technological framework can take into account multiple internal and external intervention conditions (e.g., environmental changes, hormonal fluctuations, exercise interventions, dietary control, and therapeutic measures), offering precise and dynamic health predictions and tailored guidance.

A systematic review by Shen et al. [6] demonstrated the effectiveness of digital twins in managing cancer, diabetes, multiple sclerosis, heart failure, and other diseases, supporting their transition from concept to population-wide application. Other reviews have consolidated proof-of-concept work, data-driven strategies, and implementation roadmaps for digital twins in cardiovascular disease, laying the theoretical groundwork for the field [7–9].

Previous reviews of medical digital twins have typically taken a broad perspective, encompassing diverse disease areas, and included literature from conceptual frameworks, conference abstracts, and cross-disciplinary studies. These reviews often emphasize general methodological workflows such as data integration and model calibration, which makes their larger counts of articles not directly comparable with our 31 included studies. In contrast, the present scoping review is purposefully focused on cardiovascular disease, including only original research involving cardiovascular patients or healthcare professionals, and highlighting disease-specific use cases. This focused approach allows a more relevant summary of implementation steps, application areas, and challenges. It also gives practical guidance for future research in cardiovascular digital twins.

2. Methods

2.1. Overview

This scoping review followed the Joanna Briggs Institute guidelines [10–12] and reported that it adhered to the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) checklist (Multimedia Appendix 1) [13]. The authors were trained in scoping review methodologies using the Joanna Briggs Institute Reviewers' Manual 2020 [14]. This study was not registered because the scoping review did not meet PROSPERO registration requirements.

2.2. Define the research objectives and questions

The study addressed the following questions: (1) What are the steps for leveraging digital twin technology in cardiovascular diseases? (2) What applications exist for digital twin technology in cardiovascular diseases globally? (3) What challenges are encountered in applying digital twins to cardiovascular care?

2.3. Search strategies and data sources

Search strategies were developed by referencing existing literature and in close collaboration with a medical librarian experienced in systematic reviews. A systematic electronic search was conducted across eight databases (PubMed, EBSCO, Web of Science, WILEY, China Wan-Fang Database, China National Knowledge Infrastructure, China Weipu Database, and SinoMed). The search spanned from database inception to May 2025. In PubMed, we prioritized the use of Medical Subject Headings (MeSH) to ensure precise indexing and comprehensive coverage. For records not indexed with MeSH terms, equivalent free-text terms were used to minimize omissions. Boolean operators (AND, OR) were applied to combine subject terms and keywords, and the strategy was tailored to the structure of each database. The bibliographies of relevant reviews and articles were hand-searched for potential studies for inclusion. The keywords included ‘cardiac disease’ and ‘digital twin’. The search was restricted to peer-reviewed journal articles published in English and Chinese. Other types of grey literature (e.g., conference abstracts) were excluded owing to the lack of details on the study methodology or findings. All retrieval strategies are shown in Multimedia Appendix 2.

2.4. Inclusion and exclusion criteria

The following inclusion criteria were used to select full-text articles: (1) Studies involving CVD patients or healthcare personnel; (2) Studies focusing on digital twin applications in cardiovascular care; (3) Original research (e.g., current, intervention, or case studies).

Exclusion criteria were: (1) Animal studies, abstracts, conference papers, reviews, editorials, or correspondence; (2) Studies without accessible full texts; (3) Studies not published in Chinese or English; and (4) Studies proposing digital twin frameworks without case studies.

2.5. Digital twin review definition

To clarify the boundaries between predictive modeling based on digital twins and traditional predictive models, we emphasize the key features of cardiovascular digital twins: 1) They have dynamic adaptability, capable of integrating real-time data to update simulation results, whereas traditional models are static; 2) Personalization, based on patient-specific data rather than population trends; 3) Support for bidirectional simulation for intervention testing. 4) They go beyond unidirectional prediction and integrate multi-scale data to simulate systemic interactions, unlike single-parameter traditional models.

2.6. Study selection and data extraction

Citations were imported to EndNote X9 (<https://www.myendnoteweb.com/>), and duplicates were removed. The study selection process was carried out by two investigators (ZHN and ZXL). After duplicate studies were excluded, the investigators independently assessed records based on the titles, abstracts, and full texts. Disagreements were resolved by a third investigator (WLJ). Two investigators jointly developed a data charting form using Microsoft Excel to determine which variables to extract. For each included study, the following characteristics were extracted: authors, country, study population, sample size, study design, data source, implementation steps, presentation forms, functional role, methodological details, and challenges. Additional information on each

variable was given in Multimedia Appendix 2. The extracted data were synthesised into narrative descriptions and descriptive statistics.

3. Results

3.1. Search results

A total of 545 articles were retrieved, with 83 from PubMed, 215 from Web of Science, 219 from EBSCO, 8 from WILEY, 9 from China Wanfang Database, 5 from China Knowledge Network, 3 from China Weipu Database, and 3 from Sinomed. After duplicates were removed, 463 articles remained. Title and abstract screening excluded 401 articles, leaving 62 for full-text review.

During the full-text review, 31 articles were excluded due to irrelevance, incompatible study type, or unavailable full texts. Ultimately, 31 studies were included, all published in English. The screening process is detailed in Fig. 1. Characteristics of the 31 included studies are summarised in Table 1. Reasons for document exclusion at the full-text review stage are provided in Multimedia Appendix 2. A consolidated summary of Table 1 is provided in Multimedia Appendix 2 to facilitate an integrated review.

3.2. Implementation steps of the digital twin application in cardiovascular diseases

Among the 31 included studies, the implementation of digital twins in cardiovascular diseases encompasses five phases: data acquisition, model construction and personalisation, calibration and validation,

simulation and analysis, and application for decision support or medical education. In addition, the modeling hierarchy was defined. Table 1 shows whether the whole patient, an organ system, a single organ, or a specific tissue was modeled. This helps readers understand the scope and level of detail of the model (see Fig. 2).

3.2.1. Data acquisition

Clinical data were utilised in all studies, including patient demographics, treatment records, diagnostic results (e.g., ECG, echocardiography, CT, MRI), and genetic information. Additional data sources were incorporated in 32 % of studies (10/31), including open databases (16 %, 5/31), virtual simulation platforms (13 %, 4/31), video capture [38], and wearable monitoring [41]. Collected data comprised cardiovascular anatomy (42 %, 13/31), physiological parameters (55 %, 17/31), and disease-specific features (61 %, 19/31) to support precise model construction.

3.2.2. Model construction and personalisation

Constructing personalised models is a crucial step based on the acquired data. All 31 studies employed multivariate modelling techniques. While 26 % (8/31) utilised frameworks like CircAdapt, 71 % (22/31) integrated advanced methods such as deep learning, statistical shape modelling, and generative models to account for disease complexity and patient variability.

3.2.3. Model calibration and validation

In 31 papers, the models were evaluated by comparing the calibration and validation methods. Key cardiac function indices predicted by

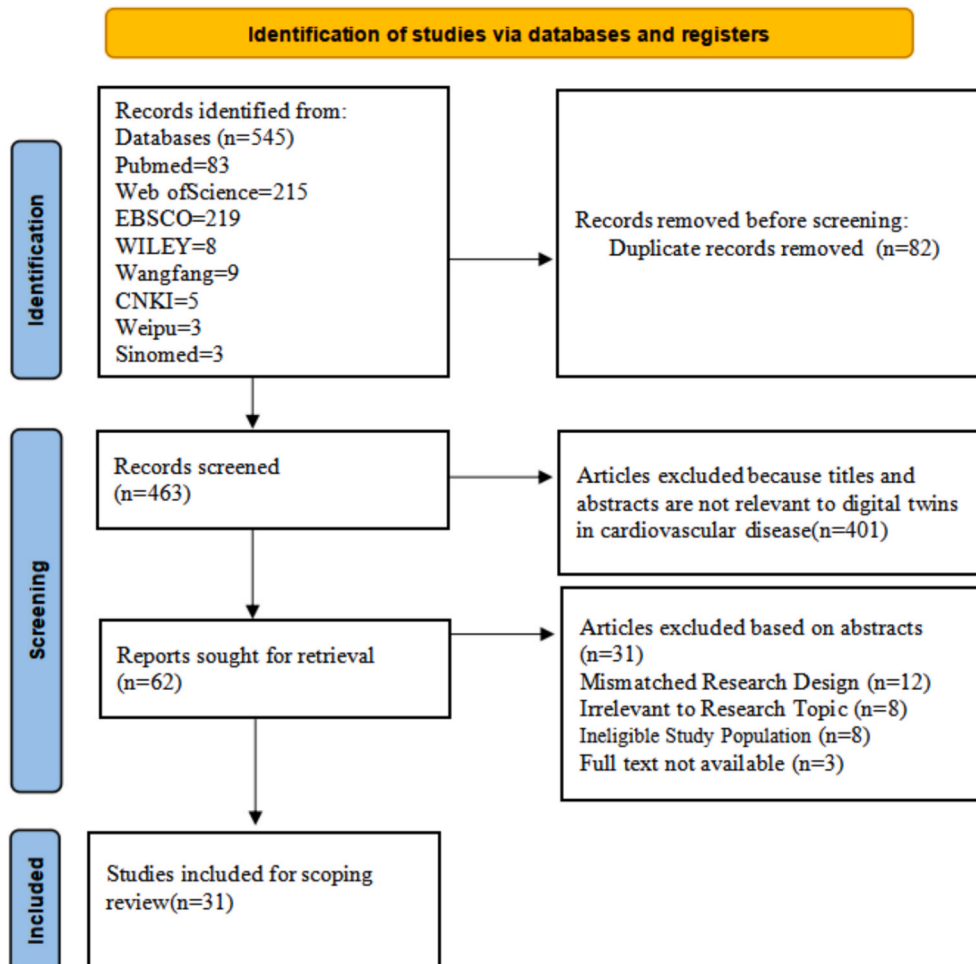


Fig. 1. PRISMA 2020 flow diagram.

Table 1
Characteristics of included studies.

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Koopsen [15], 2023	Netherlands	heart failure patients	45	Retrospective Cohort Study	Patient-specific virtual cardiac models were developed by integrating echocardiographic data into the CircAdapt cardiovascular model	Clinical data from medical institutions	Left ventricular volume, ejection fraction, myocardial strain distribution, myocardial stress, and workload differences	Data collection → construction of personalised models → virtual pacing simulation → comparison of predicted values with actual treatment outcomes	Virtual pacing simulations and prediction reports supported decisions regarding cardiac resynchronisation therapy implementation and optimisation.	Challenges in model personalisation, parameter estimation, and model complexity	PHM, PITE	OG
Ruben [16], 2022	England, Spain, China	Patients with cardiac arrhythmias	365	Validation Studies	Virtual patient model based on synthetic data	Clinical data from medical institutions and publicly available databases	Myocardial conduction velocity, anatomical characteristics of the outflow tract, and myocardial heterogeneity.	Data collection → synthetic virtual data generation → deep learning training → evaluation of classification accuracy → optimisation of model performance, combining real patient data and virtual data to train a hybrid model to improve robustness	To improve the accuracy of classification and localisation of ventricular arrhythmia sources, and support medical research and device design improvement.	Challenges in data collection, model complexity, and clinical application	PITE	OG
Van Osta[17], 2021	Netherlands, Norway	Patients with cardiac arrhythmias	There were 9 real patients and 9 virtual patients	Validation studies	A virtual pathological twin system based on personalised cardiac modelling	Clinical data from medical institutions and generated patient data using a virtual simulation platform	Local activation time, myocardial stress and stiffness, parameters involved in fibre orientation, calcium kinetics, and regional mechanical properties	Data collection → model personalisation modelling → parameter estimation and uncertainty quantification → Simulating disease progression → evaluating the predictive ability and applicability of the model for different patient groups	Individualised assessment of myocardial mechanical changes and identification of high-risk areas in disease progression can provide a basis for disease classification, diagnosis, and treatment planning.	Challenges in parameter estimation, model validation, and clinical application	PIR	OG
Kim [18], 2024	South Korea, America	Patients with abdominal aortic aneurysm and healthy controls	114 patients, plus corresponding healthy control data	Validation studies	Personalised aortic aneurysm monitoring system based on transmission line model	Clinical data from medical institutions	Abdominal aortic aneurysm diameter change, pulse travel time of blood pressure waveform, reflection coefficient and terminal impedance.	Data collection → model parameter adjustment → dynamic simulation (simulating the effect of abdominal aortic aneurysm on waveform) → result verification (compared with real data)	Personalised abdominal aortic aneurysm monitoring, postoperative efficacy evaluation, and auxiliary decision-making.	Challenges in model accuracy, parameter identification, and clinical application	PITE	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Chakshu [19], 2021	England	Patients with abdominal aortic aneurysm and healthy people	4137 virtual patients and 4522 patients	Observational Study	A virtual human circulatory system combining reverse analysis and deep learning	Clinical data from medical institutes, the virtual simulation platform, and a public database	Blood pressure, heart rate variability, abdominal aortic diameter, drug response, hydrodynamic parameters, arterial impedance parameters	Virtual patient generation (through deep learning to generate a variety of waveforms) → model training → simulation of cardiac hemodynamic characteristics → application to patient classification and prediction	To achieve non-invasive cardiovascular health monitoring, accurately predict the severity of abdominal aortic aneurysm, and provide a diagnostic basis.	Challenges in model building, model complexity, and data collection	PIR	OS
Poletti [20], 2022	America	Patients with Coronary Artery Disease	3	Validation Studies	Digital twin based on coronary CT images	Clinical data from medical institutions	Vessel diameter and lumen patency rate; Stent coverage and contact area; Blood flow shear force and pressure distribution.	Data collection → reconstruction of stent geometry model → simulation of postoperative blood flow distribution → optimisation of stent implantation plan → evaluation of postoperative recovery and improvement direction of stent design.	To improve the vascular patency rate after stent implantation. Optimisation of stent design; To improve the planning process of stent implantation and the prediction of recovery effect.	Challenges in model building, data quality, and clinical application	PIRE, OCT	OG
Li [21], 2024	England, China, Germany	Patients with Myocardial Infarction	17	Validation Studies	Personalised myocardial infarction model based on MRI and ECG	Clinical data from medical institutions and publicly available databases	QRS characteristics (onset, amplitude and duration); Myocardial fibre orientation; Volume and location of the lesion area	Data collection → Double-branch model training (injury characteristics inference) → simulation of myocardial electrical activity → verification of the agreement between simulation results and real waveforms → clinical application evaluation	Prediction of abnormal electrical activity in the early stage of myocardial infarction; QRS waveform aided diagnosis; The directional distribution of myocardial fibres was provided to support surgical planning.	Challenges in data fusion, model complexity, and clinical application	PIR	OG
Thangaraj [22], 2023	America	heart failure patients	13,912	Validation Studies	The RCT digital twin (RCT-Twin) that reflects the covariate patterns of real-world patients generated by the RCT-Twin-GAN model	Clinical data from medical institutions and Randomised Controlled Trial (RCT) data	age, left ventricular ejection fraction, heart rate, creatinine, and use or nonuse of angiotensin-converting-enzyme inhibitors, treatment assignment and	Data collection → data preprocessing → model development: employing generative adversarial networks to establish digital twin models, extract effective variables,	To simulate the effect of RCT in real-world patients and evaluate the universality of RCT. It can be used to analyse the changes in treatment effects in different patient groups and provide a	Challenges in data matching, model training, clinical application, ethics and data privacy security	PIR, OCT	OS

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
							cardiovascular outcomes	and carry out comparisons utilizing RCT data → Simulation: conduct simulations of the model to generate potential risk predictions and effect transformations → validation: verify the efficacy and accuracy of the model by comparing it with real-world data → application	reference for treatment decision-making.			
Zhang [23], 2023	America	Patients with cardiomyopathy with arrhythmia	16	Retrospective Cohort Analysis	3D patient-specific electrophysiological ventricular model	Johns Hopkins Arrhythmogenic right ventricular cardiomyopathy (ARVC) database	Age, gender, Cardiac function parameters, Genotype, Structural remodelling indicators, electrophysiological characteristics, ventricular tachycardia (VT)-related indicators	Data collection → model development (Geno-DT model) → simulation → validation (prediction of VT loop position) → application	To predict the circuit of ventricular tachycardia and help in the development and optimisation of treatment plans	Challenges in model personalisation, data collection, and clinical application	PITE	OG
Thangaraj [22], 2023	America	heart failure patients	13,912	Validation Studies	The RCT digital twin (RCT-Twin) that reflects the covariate patterns of real-world patients generated by the RCT-Twin-GAN model	Clinical data from medical institutions and Randomised Controlled Trial (RCT) data	age, left ventricular ejection fraction, heart rate, creatinine, and use or nonuse of angiotensin-converting-enzyme inhibitors, treatment assignment and cardiovascular outcomes	Data collection → data preprocessing → model development: employing generative adversarial networks to establish digital twin models, extract effective variables, and carry out comparisons utilizing RCT data → Simulation: conduct simulations of the model to generate potential risk predictions and effect transformations → validation: verify the efficacy and accuracy of the model by comparing it with real-world data → application	To simulate the effect of RCT in real-world patients and evaluate the universality of RCT. It can be used to analyse the changes in treatment effects in different patient groups and provide a reference for treatment decision-making.	Challenges in data matching, model training, clinical application, ethics and data privacy security	PIR, OCT	OS

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Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Zhang [23], 2023	America	Patients with cardiomyopathy with arrhythmia	16	Retrospective Cohort Analysis	3D patient-specific electrophysiological ventricular model	Johns Hopkins Arrhythmogenic right ventricular cardiomyopathy (ARVC) database	Age, gender, Cardiac function parameters, Genotype, Structural remodelling indicators, electrophysiological characteristics, ventricular tachycardia (VT)-related indicators	Data collection → model development (Geno-DT model) → simulation → validation (prediction of VT loop position) → application	To predict the circuit of ventricular tachycardia and help in the development and optimisation of treatment plans	Challenges in model personalisation, data collection, and clinical application	PITE	OG
Azzolin [24], 2023	Germany, Spain	Patients with Atrial Fibrillation	29	Validation Studies	A personalised computational model of the atrium	Clinical data from medical institutions	Age, gender and other basic patient information, electrophysiological indicators, fibrosis-related indicators, ablation strategy-related indicators	Data collection → mng AugmentA algorithm to generate personalised atrial model → Simulation: simulating the effects of different ablation strategies on atrial fibrillation (AF) → Validation: comparing the simulation results with the clinical situation to evaluate the success rate of different ablation strategies → application	To compare the effects of different ablation strategies on terminating AF and preventing AF recurrence, and to provide evidence for individualised ablation therapy	Challenges in model building, clinical application	PITE	OG
Koopsen [25], 2024	Netherlands	heart failure patients	13	Validation Studies	Generation of imaging-based digital twins of patients with left ventricular mechanical incongruity	Clinical data from medical institutions	Age, gender, QRS width and other basic patient information, left ventricular (LV) lumen volume, myocardial strain, myocardial work, model parameters	Data collection → model development based on the CircAdapt model → simulation of physiological and pathological processes of the patient's heart → validation: comparing the degree of fit and data consistency of the digital twin model with the patient's clinical data to verify the credibility of the model → application	To generate a patient-specific digital twin model to assist in the diagnosis and treatment decisions of patients with LV mechanical incongruity	Challenges in parameter selection, model validation, and clinical application	PHM, PITE	OG
Kirkels [26], 2023	Netherlands, Norway	Patients with cardiomyopathy with arrhythmia	82	Observational Study	Personalised patient cardiac simulation models	Clinical data from medical institutions	Age, gender, follow-up time, pathogenic gene variation, left ventricular ejection fraction, left ventricular global	Data collection → using CircAdapt model and combining patient data to construct a personalised digital	To create a digital twin model of the patient's heart, simulate the physiological state of the heart, and	Challenges in data collection, model accuracy, and clinical application	PITE	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
∞	Garot [27], 2024	France, Spain, Germany, Denmark, Canada	Patients with Atrial Fibrillation	106	Retrospective Cohort Study	3D models based on individual patient CT data	Clinical data from medical institutions	longitudinal strain, right ventricular free wall longitudinal strain, right ventricular segmental deformation parameters, right ventricular tissue characteristics	longitudinal strain, right ventricular free wall longitudinal strain, right ventricular segmental deformation parameters, right ventricular tissue characteristics	twin model → simulating cardiac physiological state by inputting actual patient data → evaluating and verifying the ability of the model to reflect disease progression → application	analyse the disease progression of patients in different age groups, to provide a basis for understanding the progression of myocardial disease in patients and assist clinical decision-making	OG
								Cardiac CT images, real-time fluoroscopic images, intraoperative transesophageal echocardiography and postoperative follow-up CT data were collected	Data collection → 3D model construction (FEops (HEARTguide™) → model and fluoroscopy image fusion → intraoperative device positioning and implantation → postoperative follow-up evaluation.	Challenges in data Fusion, model accuracy, and clinical application	OCT	
								right ventricular (RV) mechanical delay, RV wasted power ratio, RV pump work and myocardial work efficiency	Establishment of CircAdapt model → calibration and verification of digital twin → Lung valve replacement and RV-CRT simulation → Comparison of model prediction and actual intervention effects	Individualised right ventricular function assessment and treatment suggestions were provided to assist in the decision-making of right ventricular synchronisation therapy. It can also predict the possibility of improving the patient's exercise capacity, which can be used for personalised management and preoperative planning of postoperative patients.	PHM, OCT	
∞	Ložek [28], 2024	Czech Republic, Netherlands	Postoperative patients with tetralogy of Fallot	2	Validation Studies	The patient-specific cardiac and circulatory system model (CircAdapt) was used to simulate individualised right ventricular synchronisation therapy	Clinical data from medical institutions					OS
∞	Camps [29], 2024	England, Brazil, Spain	Four healthy individuals and two patients with hypertrophic cardiomyopathy	6	Retrospective Cohort Study	Multiscale ventricular model generated from individualised MR and ECG data	Clinical data from medical institutions	Purkinje network root node location, ventricular conduction velocity, regional activation time map, and QRS	Data collection → Model geometry reconstruction → activation root node inference → Purkinje network generation	It supports cardiac pathological and electrophysiological research in a normal state and realises personalised	Challenges in data fusion, model complexity, and clinical application	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
							morphology matching	→ Biophysical calibration (based on single-domain model) → validation and simulation → Comparison of the match between simulated QRS and clinical data	modelling of ventricular activation sequences for arrhythmia research and computer-assisted clinical trials.			
Yang [30], 2022	China	Congenital heart disease	3	Validation Studies	A digital model of the fetal heart based on a plaster cast and CT scan	Clinical data from medical institutions and tool-generated virtual ultrasound images	Anatomy of fetal heart chambers, main vessels, ductus arteries and vena cava and correspondence with virtual ultrasound images	Plaster casting production → CT scanning volume data generation → 3D anatomical model segmentation → Virtual ultrasound function implementation (based on PLUS tool and 3D Slicer software) → Model application in fetal ultrasound education and evaluation	To provide an intuitive relationship between fetal cardiac ultrasound anatomy and an ultrasonic section for standard section training and teaching of complex cardiac anatomy.	Challenges in data collection, model building, and clinical application	ME	OG
Qian [31], 2024	England	It covers healthy individuals and patients with cardiovascular diseases such as heart failure and bundle branch block	3464	Retrospective Cohort Study	Personalised, computational models of the heart with physical and physiological constraints built by integrating multimodal data	Large open biomedical database	Age, gender, BMI, QRS duration, Myocardial conduction velocity, biventricular myocardial mass, left ventricular ejection fraction, cardiac-related disease diagnosis, mental health-related phenotype	Data collection → anatomical model generation (mesh reconstruction) → electrophysiological model construction → calibration and validation → association analysis and functional evaluation to predict individual pathological features	Identification of cardiac electrophysiological abnormalities, disease risk assessment, and guidance of personalised treatment and drug development; To support cardiovascular health assessment in large-scale clinical studies and development of potential therapeutic targets.	Challenges in data collection, model personalisation, and model validation	PIR	OG
Romero [32], 2021	Spain, England	Patients with ascending aortic aneurysm	26 real patient data and 3000 virtual patient cohorts	Validation Studies	Virtual patient models are generated based on statistical shape modelling and machine learning	Clinical data from medical institutions	The radius, curvature, morphological parameters and tortuosity of the aortic sinus, ascending aorta and middle ascending aorta were measured.	Data collection → aortic grid modelling and feature extraction → sample generation (statistical shape modelling and generative adversarial network) → sample	Virtual trials to understand the phenotypic diversity of aortic aneurysms and support device design and drug evaluation; optimisation of sample generation algorithms and	Challenges in data matching, data collection, and model validation	OCT	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Ishwaran, [33], 2024	America	Adult patients undergoing cardiac surgery	52,792	Observational Study	Virtual patient model based on patient preoperative variables and postoperative outcomes	Clinical data from medical institutions	Postoperative mortality, stroke, deep sternal wound infection, renal failure, prolonged intubation, reoperation, prolonged hospital stay, etc. The input variables included patient age, left ventricular ejection fraction, albumin level, and disease severity.	screening and verification based on clinical standards → data analysis and application Data collection → data preprocessing → model development (random forest) → model calibration (homogeneity analysis) → results visualisation → application to quality improvement and hospital comparison.	accuracy of predictive analyses. Virtual sample generation, distribution analysis of anatomical features and cohort construction of specific pathological populations applied to clinical trials. To support surgical quality assessment and improvement, help predict postoperative risks of patients, and provide a decision-making basis for personalised treatment plans. At the same time, it enhances data understanding and operability through transparent visualisation tools.	Challenges in data quality, model complexity, and clinical application	PIR	WP
Rouhollahi [34], 2023	America	Patients with coarctation of the aorta	80	Validation studies	Patient-specific 3D aortic valve geometry model generated based on CT imaging data	Clinical data from medical institutions	Calcification volume, surface area, radial location, angular location and arc Angle of the calcification centre of mass, distribution of three valves (left, right and non-coronary) and their corresponding calcified islands	Data collection → data preprocessing → aortic root reconstruction (deep learning segmentation) → aortic valve landmark detection and geometric modelling → calcification segmentation and repositioning → result visualisation	It is used to evaluate the effect of calcification on aortic valve pathology to support Transcatheter Aortic Valve Replacement (TAVR) device design and surgical planning. To improve the predictive ability of postoperative outcomes and promote the application of personalised medicine.	Challenges in data quality, model building, and clinical application	PITE, OCT	OG
Azzolin [35], 2023	Germany	Patients with Atrial Fibrillation	29	Validation Studies	A personalised computational model of the patient's atrium	Clinical data from medical institutions	Age, gender, atrial opening labelling, fibre orientation, conduction velocity, local activation time, root mean square error of simulated and clinical local	Data collection → Construction of an atrial digital twin via AugmentA pipeline → Electrophysiological simulation → Virtual validation(13	Provide personalised atrial models to support arrhythmia susceptibility analysis, catheter ablation planning, and postoperative simulation, which	Challenges in model building, data quality clinical application	PITE	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Sakata [36], 2024	America	Patients with Atrial Fibrillation	26	Validation Studies	Personalised atrial computational model based on clinical data	Clinical data from medical institutions	activation time maps, atrial region labelling, cardiac magnetic resonance imaging-related indicators Age, gender, body mass index, left atrial/right atrial volume, left atrial/right atrial fibrosis burden, fibrosis density, fibrosis entropy, rotor position, local activation time, bipolar voltage, fractal signal area potential, clinical events	ablation strategies comparison in 29 patient models)→ Clinical translation feasibility(PersonAL plan export to mapping system) Data collection → data segmentation and fibrosis modelling → rotor localisation and classification → simulation ablation → verification of arrhythmogenic non-inducibility and reduction of ablation lesions → application	can be used for precision medicine and clinical decision support. To optimise the catheter ablation strategy, reduce the risk of inducing arrhythmia, improve the success rate of the procedure and reduce re-ablation, which can be used for preoperative evaluation and intraoperative personalised ablation guidance.	Challenges in data collection, model complexity, and clinical application	PITE	OG
Jung [37], 2022	Austria	Patients with coarctation of the aorta	7	Validation Studies	An electromechanical digital twin of the heart with biophysical detail	Clinical data from medical institutions	Cardiac cycle length, maximum and rate of change of left ventricular pressure/volume, pressure–time derivative, stroke volume, intracellular calcium concentration, fibre stretch, active stress, passive stress, electrophysiological parameters	Data collection → organ-level personalisation based on low-fidelity models → extraction of median active stress and fibre stretch rate trajectories → fitting model cellular stress and calcium kinetic parameters → application	Simulating patient-specific cardiac functional behaviour is used to evaluate the effectiveness of different treatment options, as well as to optimise decision support for afterload-related surgery.	Challenges in model personalisation, computational cost, and model validation	PITE	OG
Chakshu [38], 2019	England	Healthy volunteers and one patient with severe carotid stenosis	The number of healthy volunteers is not explicitly mentioned. One patient with a real case of severe stenosis	Validation Studies	A semi-active digital twin model where the results are presented as a curve of vibration amplitude over time, including a comparison plot of actual measured versus virtual generated vibrations.	Video (capturing head vibrations) and virtual modelling data	Age, weight, height, neck length, heart rate, cardiac output, pulse wave velocity, mean arterial pressure, percentage of carotid stenosis, head vibration amplitude, frequency	This study constructs a semi-active human digital twin, which belongs to the classification of digital twins. The core of this study is to establish a virtual representation of the degree of carotid artery stenosis by coupling computational mechanics (flow-vibration model) and computer vision (facial video analysis).	The severity of carotid artery stenosis is detected for stroke risk screening and treatment priority assessment. The patient's video was recorded by a smartphone, and the degree of carotid artery stenosis was analysed by algorithm processing and model comparison. It is suitable for remote health monitoring	Challenges in data collection, model accuracy, and clinical application	PIR	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Svensson [39], 2024	America	Adult patients undergoing cardiac surgery	6989	Retrospective Cohort Study	A virtual model based on the preoperative characteristics of the patient	Clinical data from medical institutions	Preoperative characteristics: age, BMI, Left Ventricular Ejection Fraction, surgical characteristics: proportion of coronary artery bypass grafting, type of valve surgery; Postoperative outcomes: mortality, stroke, reoperation rate, length of hospital stay	Data collection → random forest model development → risk prediction and model validation → comparison of data differences between hospitals → improvement suggestions	To optimise the surgical strategy, reduce mortality and postoperative complications, and provide data support for inter-hospital benchmarking and improvement measures	Challenges in data quality, complexity, and clinical application	OCT	WP
O'Hara [40], 2024	America, Switzerland	Patients with Myocardial Infarction	8	Validation Studies	Patient-specific myocardial models incorporating MRI data	Clinical data from medical institutions	Core scar area, grey zone volume, fibre direction, Ventricular tachycardia (VT) path induced by rapid pacing and cardiac fibre conduction characteristics	MRI acquisition → image segmentation → reconstruction of 3D myocardial model → rapid pacing simulation → prediction of VT path → verification of ablation effect and optimisation objective	To identify the origin path of ventricular tachycardia after myocardial infarction, optimise the radiofrequency catheter ablation strategy, improve the effect and reduce the risk of postoperative recurrence, and generate a personalised treatment plan	Challenges in data quality, model building, and clinical application	PITE	TS
Yang [41], 2024	China	Users of implantable cardiac devices	100 virtual patients	Validation Studies	A digital twin framework for implantable device-assisted therapy to generate patient-specific simulation models	Wearable devices and Clinical data from medical institutions	Specific arrhythmia or treatment events, including device electrical stimulation intensity, heart rate target range, patient physiological signals, etc.	Multi-source data acquisition → Dual-model construction (physiological mechanism model + state transition model) → Reinforcement learning optimisation (virtual simulation environment + Proximal Policy Optimisation algorithm) → Dynamic verification (statistical indistinguishability test + virtual patient experiment) → Model	To provide personalised treatment optimisation plans to improve the therapeutic effect of implantable devices and reduce false triggers and other side effects. For preoperative planning or intraoperative optimisation of equipment parameters, patients use the remote monitoring system to manage the	Challenges in model personalisation, clinical application, and data privacy and security challenges	PHM	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Strocchi [42], 2023	England, Austria, France	Heart failure patients	1	Validation Studies	Electromechanical model of a four-chamber heart	Clinical data from medical institutions	12-lead ECG, CT, and invasively recorded left ventricular pressure	deployment with closed-loop update (clinical parameter optimisation + weekly adaptation) Data collection → construction of four-chamber heart geometric model → simulation → validation using Gaussian process simulator → application	disease and warn of high-risk events. The study of cardiac physiological and pathophysiological mechanisms is used to understand the impact of cardiovascular diseases on cardiac function and provide insights for patient-specific treatment planning	Challenges in model complexity, computational cost, and model validation	PITE	OG
Gu [43], 2025	America	Heart failure patients	343	Validation Studies	Virtual models constructed based on patients' vital characteristics	Clinical data from a medical institution	Left/right ventricular systolic function, passive stiffness, vascular resistance, pericardial constraint, mitral/tricuspid regurgitation fraction	Data collection → Construction of cardiac digital model → Adjustment of model parameters to fit individual characteristics → Automatic classification of disease types by AI → Prediction of survival rates in different subgroups	Identify different types of conditions, predict the risk of patient death, and optimise treatment options.	Data inconsistency, lack of parameter stability, and lack of validation of longitudinal data	PIR	OG
Sakata [44], 2025	America	Patients with persistent atrial fibrillation	35	Observational study	Virtual modelling based on comparison of ablation effects	Clinical data from a medical institution	Fibrosis fraction, fibrosis entropy, low voltage zone, pulmonary vein vestibular fibrosis distribution	Data collection → Constructing a model of atrial fibrosis → Simulating the effects of different surgical protocols → Locating the root	Simulate the effect of different ablation programs to help doctors choose the most suitable treatment for patients	Model personalisation challenges, computational cost challenges, and model validation challenges	PITE	OG

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Table 1 (continued)

Author, Year	Country	Study population	Sample size (n)	Study design	Presentation forms of digital twin	Data source	Parameters	Implementation Steps	Functional role and application	Challenges	Application classification label	Modeling hierarchy Type
Waight [45], 2025	America	Patients with ventricular tachycardia in structural heart disease	18	Prospective Cohort Study	Virtual modelling of cardiac scarring	Clinical data from a medical institution	Bipolar voltage, unipolar voltage, EGM time course, deceleration zone, action potential properties	cause of abnormal electrical signals → Verifying against real surgical data Data collection → CMR reconstruction of ventricular scar model → simulate heart rate acceleration to trigger abnormalities → find the optimal ablation location → compare with actual surgical results	Predicting the key location of arrhythmia in advance, guiding doctors to accurately ablate and reduce post-procedure recurrence.	Data quality challenges, model complexity, doctors to accurately challenges, and clinical application challenges	PITE	TS

Note: PHE-Personalised health management; TITE-Precise individual treatment effect; PIR-Predicting individual risk; OCT-Optimising clinical trials; ME-Medical Education; WP-Whole patient; OS-Organ system; OG-Single organ; TS-Tissue.

models were compared with actual measurements in 65 % of studies (20/31). Model-simulated disease progression was assessed against actual patient trajectories in 36 % of studies (11/31). Consistency between model outputs and clinical observations was evaluated from multiple dimensions in 77 % of studies (24/31), analysing correlations between model parameters and patient clinical characteristics.

3.2.4. Simulation and prediction analysis

It is worth noting that it is important to clearly define the boundaries of predictive modeling that meet the conditions for digital twins. In digital twins, real-time data from physical entities must be used to drive the construction of dynamically updated virtual mappings. It is not based on historical data or static models for prediction. Simulation studies examined cardiovascular physiopathological processes. Among the 31 studies, physiological activities were simulated in 68 % of studies (21/31), pathological processes in 77 % (24/31), and treatment efficacy predictions were assessed in all studies.

3.2.5. Application for decision support or medical education

Digital twins were applied for clinical decision support or medical education in all 31 studies. Applications included personalised health management (13 %, 4/31), precise treatment effects (42 %, 13/31), individual risk prediction (26 %, 8/31), clinical trial optimisation (23 %, 7/31), and medical education (3 %, 1/31).

3.3. Digital twin technology in cardiovascular disease practice

Among the 31 studies, the application practice of digital twin in cardiovascular disease involved five aspects: personalised health management, accurate individual treatment effect, prediction of individual risk, optimisation of clinical trials, and medical education. We recognize that these categories can overlap; a single digital twin that leverages wearable data to titrate diuretics, for instance, may also forecast heart-failure risk. To maintain clarity, we assigned each article to the category that best reflects its primary objective, the resulting labels are provided in Table 1.

3.3.1. Personalised health management

In 13 % (4/31) of studies, digital twins enabled continuous monitoring and personalised management of cardiovascular health. Yang et al. [41] combined a wearable device and clinical data to construct a patient-specific digital twin for optimising implantable cardiac device parameters and implemented remote monitoring to alert to high-risk events and support personalised management. In addition, three studies used digital twins to simulate cardiac function, providing data and decision support for personalised rehabilitation and health management strategies [15,25,28].

3.3.2. Precise individual treatment effect

Digital twins were used in 42 % (13/31) of studies to refine treatment options for cardiovascular diseases and enhance precision and efficacy. Poletti et al. [20] optimised post-procedure blood flow distribution prediction, stent design, and implantation planning based on digital twins of coronary CT images, improving stenting precision and efficacy. O'Hara et al. [40,44] used digital twins to develop myocardial models from MRI data to improve ablation strategies for ventricular tachycardia, reducing recurrence rates. In addition, digital twins were also applied to arrhythmias [16,23,29,36,45], aortic valve disease [34,37], heart failure [15,25], and cardiomyopathy [26], simulating treatment effects across diverse strategies to provide individualised solutions.

3.3.3. Predicting individual risk

In 26 % (8/31) of studies, digital twins were employed to predict cardiovascular risk and guide clinical decision-making. Ishwaran [33] and Svensson et al. [39] used virtual models with preoperative and postoperative data to predict surgical risks and tailor treatment.

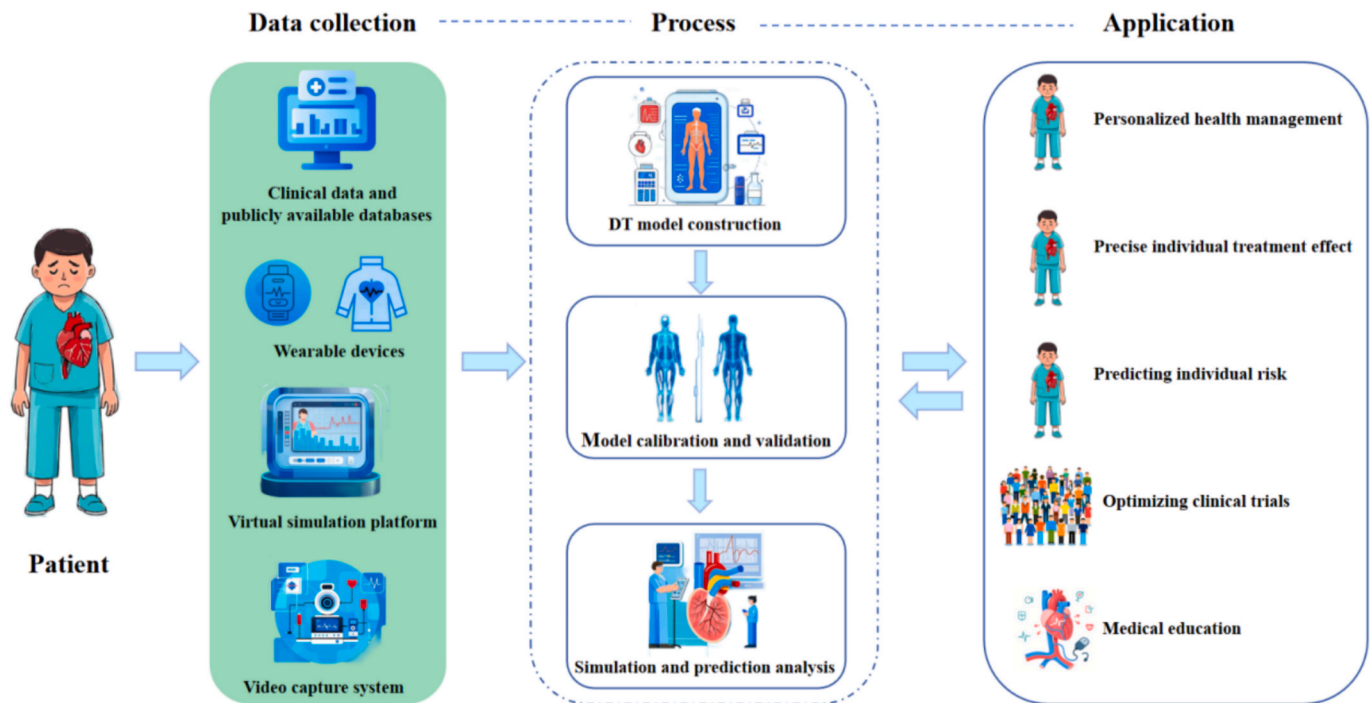


Fig. 2. Steps for constructing a digital twin in patients with cardiovascular disease.

Chakshu et al. [38] combined head vibration data with digital twins to assess carotid stenosis severity and stroke risk for screening and stratification. Employed personalised cardiac modelling to predict myocardial mechanical changes in arrhythmia [17], abdominal aortic aneurysm risk [21], and population-specific therapeutic outcomes using RCT digital twins [22]. Qian et al. [31] integrated cardiac MRI, ECG, and multimodal data to construct personalised models predicting electrophysiological abnormalities and CVD risk. Gu et al. [43] developed mechanistic computational models of the cardiovascular system to create digital twins to predict the risk of patient death and optimise treatment options.

3.3.4. Optimising clinical trials

Digital twins were used in 23 % (7/31) of studies to improve clinical trial design and evaluation. Thangaraj et al. [22] developed RCT digital twins to simulate trial effects in real-world populations, assessing result generalizability and optimising trial designs. Romero et al. [32] integrated multimodal data to construct a personalised cardiac computational model to support health assessment and therapeutic target development in large-scale clinical studies. In addition, other studies used digital twins to simulate surgical, drug, and medical device trials to assess the feasibility and effectiveness of trials in advance by simulating the trial process and results [27,28,31,34,39], reducing costs, mitigating risks, and offering innovative trial designs.

3.3.5. Medical education

Digital twins were applied for training and education in 3 % of studies (1/31). Yang et al. [30] investigated a digital twin approach using cardiovascular casts, CT digitisation, and simulation algorithms to generate virtual ultrasound images. This method provided an educational system enabling trainees to understand standard fetal echocardiography views and anatomical correlations.

3.4. Challenges of using digital twins in cardiovascular disease management

Among the 31 studies, the challenges of digital twins in CVD were categorised into three aspects: data-related, model construction and

validation, and clinical application Table 1.

3.4.1. Data challenges Table 1

Applying digital twins in CVD management presents several data-related challenges. In 26 % (8/31) of studies, data access was restricted by various factors. Data quality issues—accuracy, completeness, and transmission reliability—were noted in 26 % (8/31), affecting model performance. Data matching challenges were reported in 6 % (2/31), and data integration complexity in 16 % (5/31).

3.4.2. Model construction and validation challenges

Model personalisation and complexity were identified as challenges in 65 % of studies (20/31). Personalisation was constrained by data availability and model complexity. Training and validation shortcomings were reported in 35 % (11/31), hindering adoption due to accuracy and reliability concerns.

3.4.3. Clinical application challenges

The clinical application of digital twins faces multiple hurdles. Interpretability, clinical utility, and patient acceptance were noted in 68 % of studies (21/31). High computational costs were reported in 6 % (2/31), posing barriers in resource-limited settings. Ethical and privacy concerns were highlighted in 6 % (2/31), necessitating security measures and ethical compliance to ensure confidentiality and trust.

4. Discussion

This study evaluates digital twin technology in CVD, identifying five key construction steps and highlighting its applications in health management, treatment optimisation, risk prediction, clinical trial optimisation, and medical education. However, data limitations, model construction challenges, and barriers to clinical implementation constrain its translational potential. These findings provide insights to guide future digital twin technological developments and applications.

4.1. Digital twin construction methodology and application practice in the cardiovascular disease field

The analysis of 31 studies revealed five key steps in constructing cardiovascular digital twins: data acquisition, personalised modelling, calibration and validation, simulation analysis, and application. This framework aligns with the cardiovascular digital twin ecosystem proposed by Sel et al. [7], which is anchored in the framework defined by the U.S. National Academies of Sciences, Engineering, and Medicine. However, the reviewed studies showed limited use of wearable devices, which may compromise data integrity. Furthermore, most model validations relied heavily on static data comparisons, with little exploration of dynamic data collection, calibration, or stability. The CONNECTED project by Carbonaro et al. [46] proposed a four-layer architecture outlining data management and interoperability requirements for digital twins. The heterogeneous data management layer supports data collection, while the standards layer leverages international protocols to integrate multimodal data. The dynamic data management layer aids model construction, calibration, and intelligent application development. Finally, the application layer enables intelligent interactions and real-time feedback through a user-friendly interface, enhancing model utility in simulation and clinical decision-making.

This study highlights how digital twins, as dynamic virtual representations, enable personalised health management for cardiovascular disease, optimise treatment protocols, assist in risk prediction, improve clinical trial efficiency, and are used in medical education. These findings underscore the innovative value of digital twins in full-cycle cardiovascular disease management, consistent with prior studies [8,9]. Digital twins allow both patients and clinicians to model and predict cardiovascular outcomes by adjusting specific variables (e.g., treatment, exercise, diet, environment).

In our review, 31 studies (100 %) were dedicated to cardiovascular applications. In contrast, a recent scoping review that mapped digital twins across all healthcare decision-making found 86 studies, of which 14 (16 %) addressed diseases of the circulatory system, 12 (14 %) focused on health service utilization, and 10 (12 %) on endocrine, nutritional, and metabolic disorders [47]. This comparison highlights the relative concentration of digital twin research in cardiovascular medicine, while other fields, such as endocrinology and health services, also demonstrate substantial but smaller research efforts. A systematic review by Shen et al. [6] reported an 80 % effectiveness of digital twins in improving health outcomes across various diseases, supporting their transition from conceptual tools to broad clinical implementation. However, limitations persist, including narrow coverage of personalised health management, challenges in optimising precision therapy for complex diseases, and the need for validation of risk prediction models. Additionally, clinical trials require improved integration of virtual and real-world data to address differences in outcomes.

4.2. Implications of digital twins for nursing in cardiovascular disease

Earlier in this review, we outlined the clinical applications of digital twin technology in cardiovascular diseases. Nursing care, however, is an integral part of the diagnostic, therapeutic, and rehabilitation processes for patients with cardiovascular disease. Thus, this review further explores how digital twins can support specialized cardiovascular nursing care, offering a perspective distinct from clinical applications. The application of digital twin technology in cardiovascular disease can offer significant insights for nursing by enhancing efficiency, decision-making accuracy, and evidence translation. Digital twins optimise nursing workflows by collecting cardiovascular data through wearable devices and constructing patient-specific models. This enables caregivers to monitor patients' conditions in real time, predict potential risks, and adjust care plans promptly. Compared to traditional approaches, digital twins enable proactive and precise nursing care, enhancing care quality and patient satisfaction. Some studies have developed a personalised

computational model for patients with atrial fibrillation based on multimodal data through digital twin technology to establish a clinical decision support system for stroke prevention and aftercare [5,48]. In addition, digital twins simulate various care options, assisting caregivers in selecting optimal strategies. For example, integrating multimodal patient data and behavioural change strategies into digital twins enables the prediction of the effects of different cardiac rehabilitation training intensities and modalities on patients, ensures real-time safety monitoring, and optimises training programs [9,49]. Moreover, digital twins accelerate the generation of clinical evidence. Traditional randomised clinical trials (RCTs) are constrained by time and resources, but digital twins expedite this process [9]. By simulating diverse populations, digital twins reduce control group requirements, saving recruitment time and costs. They can also help assess the real-world applicability of RCT interventions, especially for vulnerable populations such as elderly patients with multimorbidities or those with rare diseases. By modelling treatment responses across diverse populations, digital twins reduce intervention heterogeneity and enhance clinical evidence generalizability [50].

4.3. Limitations and improvements in the application of digital twins in cardiovascular disease

This study demonstrates the promise of digital twins in cardiovascular disease while highlighting several limitations. Digital twin construction depends on accurate and comprehensive data, and deficiencies in data hinder model personalisation and accuracy, which in turn affect its application in clinical practice. These challenges align with findings from prior studies, and it is necessary to propose improvement strategies to address the key issues [4,9]. First, data relevance issues, including acquisition difficulties, quality limitations, and fusion challenges, significantly impact model reliability and accuracy. Strengthening cardiovascular data management, establishing unified standards, and leveraging technologies such as large language models to automatically process health data can improve data utilization and the accuracy of relevant devices [6]. In addition, data fusion methods should address computational complexity, reduce central integration reliance through layered processing, and improve efficiency, particularly for real-time wearable data [4]. Second, model construction and validation face limitations in personalisation and complexity, restricting their clinical adoption. Future developments should focus on creating efficient algorithms to reduce computational demands, enhance efficiency, and improve model generalisation and personalisation. This requires diversifying training data, optimising model structures, and utilising advanced technologies like cloud computing to address diverse patient needs.

Third, challenges in clinical applications include model interpretability, high computational costs, and poor integration with workflows. Embedding digital twin technology into diagnostic and treatment workflows, fostering interdisciplinary collaboration, and offering systematic training for clinical staff are crucial to improving adaptability. Ethical and data privacy concerns also require careful consideration. While ethical risks cannot be eliminated, the benefits of digital twins in prevention, treatment, and cost reduction are well recognized [5,51]. Future initiatives must address privacy, security, health equity, and social justice to maximise benefits within ethical frameworks.

It is worth noting that, compared with the studies included in Shen et al.'s [6] review, most of the research covered in this scoping review remains at the pre-clinical or proof-of-concept stage and lacks high-quality randomised controlled trials (RCTs) and multicenter validations. Thus, major bottlenecks remain in the clinical application of digital twins in cardiovascular medicine, including ethical approval, real-world effectiveness validation, and cost-effectiveness assessment. To accelerate clinical translation, future studies should prioritise the development of standardised multicenter trials, active engagement with regulatory bodies, and rigorous evaluation of cost-effectiveness.

5. Strengths and limitations

This review is the first to systematically analyse research on digital twins in cardiovascular disease, identifying key trends. However, it has some limitations. We acknowledge that restricting our search to Chinese and English databases may have underestimated the quantity and diversity of relevant evidence. This language limitation could introduce selection bias, as potentially eligible studies in other languages were not captured. Due to resource and time constraints, we did not search other country- or language-specific databases and were therefore unable to calculate the exact number of non-English, non-Chinese studies that might have been excluded. During screening of international databases, we encountered a small number of studies in other languages. However, none of these were excluded solely due to the language restriction; they were excluded for other reasons, such as study type, population, or lack of relevance to cardiovascular digital twins. From a disciplinary ecology perspective, cardiovascular digital twins represent a cutting-edge field at the intersection of medicine and engineering. Research resources are highly concentrated in Europe, America, and the Asia-Pacific region. Additionally, within the international research dissemination system, researchers in this field, even those from non-English-speaking countries, generally choose English as the language for publishing their findings to overcome regional academic barriers. Combined with the relatively low coverage of non-English academic papers in mainstream databases, these factors collectively resulted in the absence of literature related to cardiovascular digital twins in other languages in this study's search. This indirectly reflects the existence of certain regional disparities in the development of this field. Excluding conference papers may have omitted recent or preliminary findings, potentially compromising the comprehensiveness of this review. Additionally, the small sample size of 31 studies may limit the stability and generalizability of findings. Despite rigorous methods, these limitations may influence the results. Future studies with larger samples and advanced clustering methods could enhance reliability and generalizability.

6. Conclusion

This review evaluated digital twin technology in cardiovascular care, outlined five construction phases, and assessed its potential in health management, precision therapy, risk prediction, trial optimisation, and education. Although the digital twin holds great promise in the field of cardiovascular disease, its widespread adoption and practical implementation are still in the early stages. Challenges such as data quality, model complexity, clinical applicability, and privacy concerns remain significant. Addressing these issues requires improved data sharing, algorithm refinement, enhanced model generalisation, interdisciplinary collaboration, and standardised multicenter trials. Such efforts can broaden digital twins' role in advancing precision cardiology across diverse healthcare systems.

7. Data availability

Data related to the studies reported in this article are available with the corresponding author.

8. Ethics and dissemination

No ethical approval is required for the scoping review, as data will be obtained from publicly available sources. The results of this scoping review will be published in a peer-reviewed journal and presented at conferences.

CRediT authorship contribution statement

Huina Zou: Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation,

Conceptualization. **Xinglin Zheng:** Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Linjing Wu:** Writing – review & editing, Validation, Formal analysis, Data curation, Conceptualization. **Shujie Zhang:** Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization. **Polun Chang:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization. **Yuan Chen:** Writing – original draft, Validation, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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